



Linear algebra notes

Engineering mathematics (Visvesvaraya Technological University)



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Linear Algebra - I

Defn's on matrices:-

→ A set of ^{Elements} ~~mn~~ numbers arranged in m rows and n columns is called an $m \times n$ matrix (or) matrix of order $m \times n$

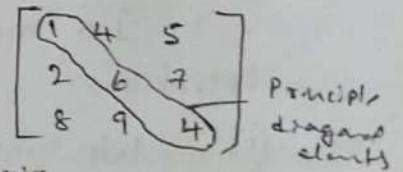
Denoted by $A = a_{ij} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$

→ The matrix having only one row is called row matrix (or) row vector

→ The matrix having only one column is called column matrix (or) column vector

→ An $m \times n$ matrix in which $m=n$ is called a square matrix of order n .

→ In a square matrix the elements from left top to the right bottom are called Principle diagonal elements (or) diagonal elements & all other elements are called non-diagonal elements.



→ In any matrix if all the elements are zero then that matrix is known as zero matrix (or) null matrix.

→ In a square matrix if except Principle diagonal elements if all the elements are zero then it is a diagonal matrix of order n .

→ In a diagonal matrix if all the elements of Principle diagonal elements are equal to one then it is a unit matrix

→ In a square matrix if all the elements below the P.D.E (Principle diagonal elements) are zero then it is an upper triangular matrix.

→ In a square matrix if all the elements above the diagonal elements are zero then it is an lower triangular matrix.

→ The transpose of a matrix A is obtained by interchanging the rows & columns of the matrix A and it is denoted by

$$A^T = [a_{ij}]^T = [a_{ji}]$$

→ A Square matrix A is said to be Symmetric matrix

if $A^T = A$ (i) $[a_{ji}] = [a_{ij}]$

→ A Square matrix A is said to be Skew Symmetric matrix

if $A^T = -A$ (ii) $[a_{ji}] = -[a_{ij}]$

→ A Square matrix A is said to be a Singular matrix

if $|A| = 0$, otherwise it is non-singular matrix.

→ If A & B are matrices $\Rightarrow$ The product AB exists, then $B^T A^T$

exists and $(AB)^T = B^T A^T$

→ $A \cdot A^T = A^T A = I$ Also $(A^{-1})^{-1} = A$

→ $A^{-1} = \frac{\text{adj } A}{|A|}$ (where $|A| \neq 0$)

[adj A is nothing but
co-factor
matrix of A
& taking transpose]
 $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 1 & 2 & 3 \end{vmatrix}$
cof of 1 is $\begin{vmatrix} 5 & 6 \\ 2 & 3 \end{vmatrix}$

-; Elementary transformation (i) Elementary Operation:-

In many matrix computations we perform the following operations

- (1) We interchange two rows (i) two columns
- (2) We multiply all elements of a row (or a column) by a non-zero constant
- (3) To k times the elements of a row (or a column) we add p times the corresponding elements of other row (or column), where k & p are non zero constants

These operations are called elementary operations (i) transformation

An elementary operation applied to rows only is called a row operation & an elementary operation applied to columns only is called column operation.

If $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$

(1) $R_2 \leftrightarrow R_3 = \begin{bmatrix} a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$

(2) $KR_2 = \begin{bmatrix} a_1 & a_2 & a_3 \\ kb_1 & kb_2 & kb_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$

(3) $R_2 \rightarrow KR_2 + PR_3 = \begin{bmatrix} a_1 & a_2 & a_3 \\ kb_1 + pc_1 & kb_2 + pc_2 & kb_3 + pc_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$

Echelon form of a matrix :-

(2)

A non zero matrix 'A' is said to be in Row Echelon Form if the following conditions satisfy

- (1) All the ~~non~~ zero rows are below ^{non} zero rows
- (2) The first non-zero entry in any non-zero row is 1
[for convenience]
The entries below 1 in the same column are zero & below the Principal diagonal elements are zero]

Ex :-

$$\begin{bmatrix} 1 & 3 & 2 & 4 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (1) \quad \begin{bmatrix} 1 & 3 & 2 & 4 \\ 0 & 1 & 2 & 9 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2) \quad \begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (3)$$

are in echelon form

→ Normal form (iv) Canonical form of a matrix

The given matrix 'A' is reduced to an echelon form first by applying a series of elementary row transformations.

Later column transformations are performed to reduce the matrix to one of the following four forms, called the Normal form of 'A'.

(1) I_r (2) $[I_r, 0]$ (3) $\begin{bmatrix} I_r \\ 0 \end{bmatrix}$ (4) $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$

where I_r is the identity matrix of order 'r'.

Ex's : (1) $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(2) $[I_3, 0] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, $[I_2, 0] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

(3) $\begin{bmatrix} I_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ $\begin{bmatrix} I_3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Exs

③

Reduce the following matrix to the Echelon form

① $A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix}$

Soln:- Given $A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix}$ $R_2 \rightarrow R_2 - 2R_1$
 $R_3 \rightarrow R_3 - R_1$

$A \sim \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & -14 & 16 \end{bmatrix}$ $R_3 \rightarrow 7R_3 - 14R_2 \Rightarrow A \sim \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix}$ $R_2 \rightarrow -\frac{1}{7}R_2$

$\therefore A \sim \begin{bmatrix} 1 & 3 & -2 \\ 0 & 1 & 8/7 \\ 0 & 0 & 0 \end{bmatrix}$ is the row echelon form of 'A'

② $A = \begin{bmatrix} 2 & -1 & 1 \\ 3 & -1 & 1 \\ 4 & -1 & 2 \\ -1 & 1 & -1 \end{bmatrix}$ $R_1 \leftrightarrow R_4 \Rightarrow A \sim \begin{bmatrix} -1 & 1 & -1 \\ 3 & -1 & 1 \\ 4 & -1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$ $R_1 \rightarrow -R_1$

$A \sim \begin{bmatrix} 1 & -1 & 1 \\ 3 & -1 & 1 \\ 4 & -1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$ $R_2 \rightarrow R_2 - 3R_1$
 $R_3 \rightarrow R_3 - 4R_1$
 $R_4 \rightarrow R_4 - 2R_1 \Rightarrow A \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -2 \\ 0 & 3 & -2 \\ 0 & 1 & -1 \end{bmatrix}$ $R_3 \rightarrow 2R_3 - 3R_2$
 $R_4 \rightarrow 2R_4 - R_2$

$= A \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ $R_2 \rightarrow \frac{1}{2}R_2$
 $R_3 \rightarrow \frac{1}{2}R_3 \Rightarrow A \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ is the row echelon form of 'A'

③ $A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & 8 \end{bmatrix}$ $R_2 \rightarrow 2R_2 + 2R_1$
 $R_3 \rightarrow R_3 - R_1 \Rightarrow A \sim \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 8 & 5 & 0 \\ 0 & -2 & 1 & 8 \end{bmatrix}$ $R_3 \rightarrow 4R_3 + R_2$

$A \sim \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 8 & 5 & 0 \\ 0 & 0 & 9 & 32 \end{bmatrix}$ $R_2 \rightarrow \frac{1}{8}R_2$
 $R_1 \rightarrow \frac{1}{9}R_3 \Rightarrow A \sim \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 5/8 & 0 \\ 0 & 0 & 1 & 32/9 \end{bmatrix}$

is the row echelon form of 'A'

Using Elementary transformation reduce the following matrices to the normal form.

① $A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 3 & 4 & 1 & 2 \\ -2 & 3 & 2 & 5 \end{bmatrix}$ $R_2 \rightarrow R_2 - 3R_1$
 $R_3 \rightarrow R_3 + 2R_1$

$$A \sim \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & -2 & 1 & 5 \\ 0 & 7 & 2 & 3 \end{bmatrix} \xrightarrow{R_3 \rightarrow 2R_3 + 7R_2} A \sim \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & -2 & 1 & 5 \\ 0 & 0 & 11 & 41 \end{bmatrix} \xrightarrow{C_2 \rightarrow -\frac{1}{2}C_2}$$

$$\Rightarrow A \sim \begin{bmatrix} 1 & -1 & 0 & -1 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 11 & 41 \end{bmatrix} \xrightarrow{\begin{matrix} C_2 \rightarrow C_2 + C_1 \\ C_4 \rightarrow C_4 + C_1 \end{matrix}} \Rightarrow A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 11 & 41 \end{bmatrix} \xrightarrow{\begin{matrix} C_3 \rightarrow C_3 - C_2 \\ C_4 \rightarrow C_4 - 5C_2 \end{matrix}}$$

$$= A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 11 & 41 \end{bmatrix} \xrightarrow{\begin{matrix} R_3 \rightarrow \frac{1}{11}R_3 \\ C_4 \rightarrow \frac{1}{41}C_4 \end{matrix}} \Rightarrow A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{C_4 \rightarrow C_4 - C_3}$$

$$= A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = [I_3, 0] \text{ is the normal form of } A$$

② $A = \begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}$ $C_2 \leftrightarrow C_1 \Rightarrow A \sim \begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & 4 & 2 & 6 \\ 1 & 2 & 3 & 1 \end{bmatrix}$ $R_3 \rightarrow R_3 - R_1$

$$= A \sim \begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & 4 & 2 & 6 \\ 0 & 2 & 1 & 3 \end{bmatrix} \xrightarrow{R_2 \rightarrow \frac{R_2}{2}} \Rightarrow A \sim \begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & 2 & 1 & 3 \\ 0 & 2 & 1 & 3 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2}$$

$$= A \sim \begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\begin{matrix} C_3 \rightarrow C_3 - 2C_1 \\ C_4 \rightarrow C_4 + 2C_1 \end{matrix}} \Rightarrow A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\begin{matrix} C_3 \rightarrow 2C_3 - C_2 \\ C_4 \rightarrow 2C_4 - 3C_2 \end{matrix}}$$

$$\Rightarrow A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \Rightarrow A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$$

Also
First
 $R_1 \leftrightarrow R_3$

$\Rightarrow$ is the normal form of A

Rank of a matrix :-

The rank of a matrix 'A' in echelon form is equal to the no of non-zero rows. It is denoted by $\rho(A)$

Any matrix of order $m \times n$ can be reduced to one of the normal forms i.e.

$$(1) I_r \quad (2) [I_r, 0] \quad (3) \begin{bmatrix} I_r \\ 0 \end{bmatrix} \quad (4) \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

In the above normal form of matrices will have 'r' non-zero rows

Hence $\rho(A) = r$

Ex's :- Find the rank of the following matrices by applying elementary ~~row~~ operations.

$$(1) \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix} \begin{matrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{matrix} \Rightarrow A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix} \begin{matrix} R_3 \rightarrow R_3 - R_2 \end{matrix}$$

$$\Rightarrow A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_2 \rightarrow \frac{1}{2}R_2 \end{matrix} \Rightarrow A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix 'A' is in the echelon form is having 2 non-zero rows.

Hence the rank of A is 2

Thus $\rho(A) = 2$

② $A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix} \quad R_1 \leftrightarrow R_2$

$A = \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix} \quad R_3 \rightarrow R_3 - 2R_1 \Rightarrow A \sim \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 2 & 3 & 4 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$

$A \sim \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} R_1 \rightarrow \frac{1}{2}R_1 \\ R_2 \rightarrow \frac{1}{2}R_2 \end{matrix} \Rightarrow A \sim \begin{bmatrix} 1 & \frac{3}{2} & \frac{5}{2} & 2 \\ 0 & 1 & \frac{3}{2} & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\therefore \rho(A) = 2$

③ $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ -2 & -3 & 1 & 2 \\ -3 & -4 & 5 & 8 \\ 1 & 3 & 10 & 14 \end{bmatrix} \quad \begin{matrix} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 + 3R_1 \\ R_4 \rightarrow R_4 - R_1 \end{matrix} \Rightarrow A$

$A \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 7 & 10 \\ 0 & 2 & 14 & 20 \\ 0 & 1 & 7 & 10 \end{bmatrix} \quad \begin{matrix} R_3 \rightarrow R_3 - 2R_2 \\ R_4 \rightarrow R_4 - R_2 \end{matrix}$

$A \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 7 & 10 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \therefore \rho(A) = 2$

④ $A = \begin{bmatrix} 2 & 1 & 3 & 5 \\ 4 & 2 & 1 & 3 \\ 8 & 4 & 7 & 13 \\ 8 & 4 & -3 & -1 \end{bmatrix} \quad \therefore \rho(A) = 2$

⑤ $\begin{matrix} ** \\ \begin{bmatrix} a_1 & a_2 & a_3 & a_4 & a_5 \\ a_2 & a_3 & a_4 & a_5 & a_6 \\ a_3 & a_4 & a_5 & a_6 & a_7 \\ a_4 & a_5 & a_6 & a_7 & a_8 \\ a_5 & a_6 & a_7 & a_8 & a_9 \end{bmatrix} \end{matrix} \quad \begin{matrix} R_2: a_1R_2 - a_2R_1 \\ R_3: a_1R_3 - a_3R_1 \\ R_4: a_1R_4 - a_4R_1 \\ R_5: a_1R_5 - a_5R_1 \end{matrix} \Rightarrow \begin{bmatrix} a_1 & a_2 & a_3 & a_4 & a_5 \\ 0 & -1 & -2 & -3 & -4 \\ 0 & -2 & -4 & -6 & -8 \\ 0 & -3 & -6 & -9 & -12 \\ 0 & -4 & -8 & -12 & -16 \end{bmatrix} \quad \begin{matrix} R_2: -R_2 \\ R_3: -2 \\ R_4: -3 \\ R_5: -4 \end{matrix}$

$\rho(A) = 2$

$$(4) A = \begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix} \quad R_1 \leftrightarrow R_4 \Rightarrow A \sim \begin{bmatrix} 2 & 3 & 1 & 4 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 4 & 0 & 2 & 1 \end{bmatrix}$$

$$\begin{aligned} R_2 &\rightarrow R_2 - R_1 \\ R_3 &\rightarrow R_3 - R_1 \\ R_4 &\rightarrow R_4 - 2R_1 \end{aligned}$$

$$\Rightarrow A \sim \begin{bmatrix} 2 & 3 & 1 & 4 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & 3 & 3 \\ 0 & -6 & 0 & -7 \end{bmatrix} \quad \begin{aligned} R_2 &\rightarrow \frac{R_2}{-2} \\ R_3 &\rightarrow \frac{1}{3} R_3 \\ R_4 &\rightarrow -R_4 \end{aligned} \Rightarrow A \sim \begin{bmatrix} 2 & 3 & 1 & 4 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 6 & 0 & 7 \end{bmatrix}$$

$$\begin{aligned} R_1 &\rightarrow \frac{1}{2} R_1 \\ R_4 &\rightarrow R_4 - 6R_2 \end{aligned}$$

$$\Rightarrow A \sim \begin{bmatrix} 1 & 3/2 & 1/2 & 2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 6 & 7 \end{bmatrix} \quad R_4 \rightarrow R_4 - 6R_3 \Rightarrow A \sim \begin{bmatrix} 1 & 3/2 & 1/2 & 2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \underline{\underline{f(A) = 4}}$$

→ Find the rank of the following matrices by reducing them to normal form.

$$(1) A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix} \quad R_1 \rightarrow R_2 \Rightarrow A \sim \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$A \sim \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 2 & 3 & 4 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2 \Rightarrow A \sim \begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} C_2 &\rightarrow 2C_2 - 3C_1 \\ C_3 &\rightarrow 2C_3 - 5C_1 \\ C_4 &\rightarrow 2C_4 - 4C_1 \end{aligned}$$

$$\Rightarrow A \sim \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 4 & 6 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{aligned} R_1 &\rightarrow \frac{1}{2} R_1 \\ R_2 &\rightarrow \frac{1}{2} R_2 \end{aligned} \Rightarrow A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} C_3 &\rightarrow 2C_3 - 3C_2 \\ C_4 &\rightarrow 2C_4 - 4C_2 \end{aligned}$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \frac{1}{2} R_2 \Rightarrow A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\therefore \underline{\underline{f(A) = 2}}$$

only try

$$A = \begin{bmatrix} 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \\ 0 & 1 & 2 & -2 \end{bmatrix} \quad ; \quad A = \begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix}$$

-: Consistency of a system of linear equations :-

A system of equations in which all the unknown quantities appears in the first degree only (i.e. coeff of x, y, z or x_1, x_2, x_3 only not x^2, y^2, z^2) is called a system of linear equations.

Consider a system of m linear equations in ' n ' unknowns as,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned} \quad \text{where } a_{ij}'\text{'s \& } b_i'\text{'s are constants.}$$

If $b_1, b_2, \dots, b_m$ are zero, then the system is said to be homogeneous (i.e. $RHS = 0$)

The set of values obtained for $x_1, x_2, \dots, x_n$ is called as the solution of the system of equations.

A system of linear equations is said to be consistent if it possesses a solution otherwise it is inconsistent.

The above system of equations can be written in the matrix form as $AX = B$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m \times 1}$$

In Particular If $x_1 = x_2 = x_3 = \dots = x_n = 0$ is a Soln of the system of homogeneous eqns & is called a trivial solution.

If atleast one x_i where $i = 1, 2, \dots, n$ is not equal to zero then it is called non-trivial solution

∴ Condition for consistency and type of solution :-

Consider a system of m eqns in n unknowns represented in the matrix form $AX = B$

Now consider a new matrix comprising of all the elements of co-efficient matrix 'A' along with the elements of the column matrix B. Such a matrix denoted by $[A : B]$ is called the 'Augmented matrix'

$$\text{i.e. } [A : B] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} : b_1 \\ a_{21} & a_{22} & \dots & a_{2n} : b_2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} : b_m \end{bmatrix}$$

The system of equations represented by the matrix equation $AX = B$ is consistent if $\rho(A) = \rho[A : B]$

Suppose $\rho(A) = \rho[A : B] = r$ (say) then the condition for various types of solution are.

1) Unique Soln:- $\rho(A) = \rho[A : B] = r = n$, where ' n ' is the no of unknowns.

2) Infinite Soln:- $\rho(A) = \rho[A : B] = r < n$

In this case $(n-r)$ unknowns can take arbitrary values.

Note: If $\rho(A) \neq \rho[A : B]$ then the system is inconsistent
(i.e. does not possess a Soln)

Working rule:-

- First we form the augmented matrix $[A:B]$ where A is co-eff matrix.
- we reduce the matrix $[A:B]$ to an echelon form by elementary row operations. This will help us to find the rank of A and also $[A:B]$ with this result we can decide the consistency of the system of equations.
- The echelon form of $[A:B]$ is changed to equation form in order to find the soln.

EXS:- Test for consistency and solve.

$$3x + y + 2z = 3$$

$$2x - 3y - z = -3$$

$$x + 2y + z = 4$$

Soln:- The augmented matrix is given by

$$[A:B] = \begin{bmatrix} 3 & 1 & 2 & : & 3 \\ 2 & -3 & -1 & : & -3 \\ 1 & 2 & 1 & : & 4 \end{bmatrix} \quad R_1 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & : & 4 \\ 2 & -3 & -1 & : & -3 \\ 3 & 1 & 2 & : & 3 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \quad -63 + 55$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & : & 4 \\ 0 & -7 & -3 & : & -11 \\ 0 & -5 & -1 & : & -9 \end{bmatrix} \quad R_3 \rightarrow 7R_3 - 5R_2 \Rightarrow \sim \begin{bmatrix} 1 & 2 & 1 & : & 4 \\ 0 & -7 & -3 & : & -11 \\ 0 & 0 & 8 & : & -8 \end{bmatrix}$$

∴ Both A & $[A:B]$ matrices have 3 non-zero rows.

$$\therefore \rho(A) = 3 \quad \& \quad \rho[A:B] = 3 \quad \text{i.e. } r = 3$$

& no of unknown $n = 3$

∴ $\rho(A) = \rho[A:B] = 3$ ($n = r = 3$) the given system of equation is consistent and will have unique soln

The above matrix can be written in the equation form as

$$x + 2y + z = 4$$

$$7y + 3z = 11$$

$$8z = -8 \Rightarrow \boxed{z = -1}$$

$$\textcircled{2} \Rightarrow 7y = 11 - 3z \quad \textcircled{3} \Rightarrow x = 4 - 2y - z$$

$$= 11 + 3$$

$$= 4 - 4 + 1$$

$$7y = 14$$

$$y = 2$$

$$\underline{x = 1}$$

Solving the above equation we get $x = 1$, $y = 2$, $z = -1$

→ Wkly Test for consistency & solve.

$$x + 2y + 3z = 14$$

$$4x + 5y + 7z = 14$$

①

$$x + y + z = 6$$

$$x - y + 2z = 5$$

$$3x + y + z = 8$$

$$\left. \begin{array}{l} x + y + z = 6 \\ x - y + 2z = 5 \end{array} \right\} n = r = 3$$

$$\underline{x = 1, y = 2, z = 3}$$

②

$$x + y + z = -3$$

$$3x + y - 2z = -2$$

$$2x + 4y + 7z = 7$$

$$\left. \begin{array}{l} x + y + z = -3 \\ 3x + y - 2z = -2 \end{array} \right\} n = r = 2, \quad \rho(A:B) = 3$$

System is not consistent

②

$$x + 2y + 3z = 1$$

$$2x + 3y + 8z = 2$$

$$x + y + z = 3$$

$$\left. \begin{array}{l} x + 2y + 3z = 1 \\ 2x + 3y + 8z = 2 \end{array} \right\} n = r = 3$$

$$\underline{x = 9/2, y = -1, z = -1/2}$$

Test for consistency and solve.

$$x + 2y + 2z = 2$$

$$3x - 2y - z = 5$$

$$2x - 5y + 3z = -4$$

$$x + 4y + 6z = 0$$

Here no. of unknowns $n=3$

Soln:

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 3 & -2 & -1 & 5 \\ 2 & -5 & 3 & -4 \\ 1 & 4 & 6 & 0 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 2R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$$

$$\begin{array}{l} 8 - 6 = -5 \\ 6 - 9 = -3 \end{array}$$

$$[A:B] \sim \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & -8 & -7 & -1 \\ 0 & -9 & -1 & -8 \\ 0 & 2 & 4 & -2 \end{array} \right] \begin{array}{l} R_2 \rightarrow -R_2 \\ R_3 \rightarrow -R_3 \\ R_4 \rightarrow \frac{1}{2}R_4 \end{array} \Rightarrow \sim \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 8 & 7 & 1 \\ 0 & 9 & 1 & 8 \\ 0 & 1 & 2 & -1 \end{array} \right] \begin{array}{l} R_3 \rightarrow 8R_3 - 9R_2 \\ R_4 \rightarrow 8R_4 - R_2 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 8 & 7 & 1 \\ 0 & 0 & -55 & 55 \\ 0 & 0 & 9 & -9 \end{array} \right] \begin{array}{l} R_3 \xrightarrow{55} \\ R_4 \xrightarrow{9} \end{array} \sim \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 8 & 7 & 1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -1 & -1 \end{array} \right] \begin{array}{l} R_4 \rightarrow R_4 - R_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 8 & 7 & 1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

System of eqn becomes

$$x + 2y + 2z = 2$$

$$8y + 7z = 1$$

$$z = -1$$

$$\therefore \rho(A) = \rho[A:B] = 3 \quad (n=r=3)$$

$$\therefore \underline{\underline{x=2, y=1, z=-1}}$$

ully :-

$$3x + 3y + 2z = 1$$

$$x + 2y = 4$$

$$10y + 3z = -2$$

$$2x - 3y - z = 5$$

→ Test for consistency & Solve.

$$x + 2y + 3z = 14$$

$$4x + 5y + 7z = 35 \quad n=3$$

$$3x + 3y + 4z = 21$$

Soln: $[A:B] = \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 4 & 5 & 7 & : & 35 \\ 3 & 3 & 4 & : & 21 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 4R_1, R_3 \rightarrow R_3 - 3R_1} \Rightarrow N \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 0 & -3 & -5 & : & -21 \\ 0 & -3 & -5 & : & -21 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2}$

$$A:B \sim \begin{bmatrix} 1 & 2 & 3 & : & 14 \\ 0 & -3 & -5 & : & -21 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$\therefore \rho(A) = \rho[A:B] = 2 = r \quad \& \quad n=3$$

$$\rho(A) = \rho[A:B] = 2 < 3 \quad (\text{i.e. } r < n)$$

$\therefore$ The system is consistent and will have infinite solutions

$\therefore$ The no of unknowns that can be chosen arbitrarily is $n-r = 3-2 = 1$

$\therefore$ The system of eqn becomes $x + y + 3z = 14$
 $3y + 5z = 21$

Let $z = k$ be arbitrary (i) $y = k$ also can be taken.

Solving the above eqn we have, $x = 7 - \frac{4k}{3}$, $y = 7 - \frac{5k}{3}$, $z = k$ represents infinite soln's $\because k$ is arbitrary

→ (2) $\begin{cases} x - y + z = 2 \\ 3x - y + 2z = -6 \\ 3x + y + z = -18 \end{cases} \quad \rho(A) = \rho[A:B] = 2 < 3$
 $\therefore n-r = 3-2 = 1$ arbitrary
 $\therefore x = \frac{-4-k}{2}$, $y = \frac{k}{2} - 6$, $z = k$

(3) $\begin{cases} x + 2y + 2z = 1 \\ 2x + y + z = 2 \\ 3x + 2y + 2z = 3 \\ y + z = 0 \end{cases} \quad \begin{matrix} n=3 \\ \rho(A) = \rho[A:B] = 2 < 3 \\ z=k, y=-k, x=1 \end{matrix}$

(4) $\begin{cases} 5x + 3y + 7z = 4 \\ 3x + 26y + 2z = 9 \\ 7x + 2y + 10z = 5 \end{cases} \quad z=k, y = \frac{1}{11}(k+3), z = \frac{1}{11}(7-16k)$

Test for consistency & solve.

$$x + y + z = -3$$

$$3x + y - 2z = -2$$

$$2x + 4y + 7z = 7$$

Soln: $[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 3 & 1 & -2 & : & -2 \\ 2 & 4 & 7 & : & 7 \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 2R_1}} \sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & 7 \\ 0 & 2 & 5 & : & 13 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & 7 \\ 0 & 0 & 0 & : & 20 \end{bmatrix}$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & 7 \\ 0 & 0 & 0 & : & 20 \end{bmatrix}$$

Here we have $\rho(A) = 2$ $\rho[A:B] = 3$

$\therefore \rho(A) \neq \rho[A:B]$ the system is inconsistent

u.l.g. $\rightarrow \begin{cases} 5x + 3y + 7z = 5 \\ 3x + 26y + 2z = 9 \\ 7x + 2y + 10z = 5 \end{cases} \quad \begin{cases} x - 4y + 7z = 14 \\ 3x + 8y - 2z = 13 \\ 7x - 8y + 26z = 5 \end{cases}$

$\rightarrow$ Investigate the values of λ & μ of the system of equations

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

may have.

(a) Unique soln

(b) Infinite soln.

(c) No soln

other ex

$$x + 2y + 3z = 6$$

$$x + 3y + 5z = 9$$

$$2x + 5y + \lambda z = \mu$$

Soln: $[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 3 & : & 10 \\ 1 & 2 & \lambda & : & \mu \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1}} \sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 1 & (\lambda-1) & : & (\mu-6) \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & (\lambda-3) & : & (\mu-10) \end{bmatrix}$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & \lambda-3 & : & (\mu-10) \end{bmatrix}$$

$$\lambda-1-2$$

$$\lambda-3$$

$$\mu-10-4$$

(a) Unique soln:- we must have $\rho(A) = \rho[A:B] = 3$.

This is possible if $\lambda-3 \neq 0$ (a) $\lambda \neq 3$

$\therefore$ The system is unique soln if $\lambda \neq 3$

(b) Infinite soln: If here we have no of unknown $n=3$

we need $\rho(A) = \rho[A:B] = 2 < 3$ This is possible if the last row of $[A:B]$ is completely zero i.e. $\lambda-3=0$ & $\mu-10=0$.

$\therefore$ The system will have infinite soln if $\lambda=3$ & $\mu=10$

(c) No soln:- here we must have $\rho(A) \neq \rho[A:B]$ This is possible if $\lambda-3=0$ & $\mu-10 \neq 0$

i.e. If $\lambda=3$ we have $\rho(A)=2$ & $\rho[A:B]=3$ if $\mu-10 \neq 0$

$\therefore$ The system is inconsistent (c) no soln if $\lambda=3$ & $\mu \neq 10$

Find for what values of 'k' the system of equations

$$x+y+z=1$$

$$x+2y+4z=k$$

$$x+4y+10z=k^2$$

Possess a solution. Solve completely in each case

Soln: $[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 1 & 2 & 4 & : & k \\ 1 & 4 & 10 & : & k^2 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - R_1} \sim \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & 1 & 3 & : & k-1 \\ 0 & 3 & 9 & : & k^2-1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 3R_2}$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 1 \\ 0 & 1 & 3 & : & k-1 \\ 0 & 0 & 0 & : & k^2-3k+2 \end{bmatrix}$$

$$\therefore \rho(A) = 2$$

For the system to be consistent, we must

have $\rho[A:B] = 2$ This is possible if $k^2-3k+2=0$

$$\therefore k=1, k=2$$

Hence we conclude that the system possesses the soln if $k=1, 2$

Further $\because \rho(A) = \rho[A:B] = 2 < 3$ ($i.e. r < n$)

Here the system will have infinite soln for $k=1, 2$

case (i) $\therefore$ Let $k=1$, Then the eqn becomes.

$$x+y+z=1$$

$$y+3z=0$$

Let $z = k_1$ be arbitrary $\therefore$ $y = -3k_1$, $x = 1+2k_1$

case (ii) $\therefore$ Let $k=2$, Then the eqn becomes.

$$x+y+z=1$$

$$y+3z=1$$

Let $z = k_2$ be arbitrary $\therefore$ $y = 1-3k_2$, $x = 2k_2$

Thus $(x = \frac{1+2k_1}{k_1}, y = -3k_1, z = k_1)$ & $(x = 2k_2, y = 1-3k_2, z = k_2)$ gives

all the soln of the given system of eqns

Solution of linear homo eqns (RHS = 0)

Let $AX=0$ represents m linear eqns in n unknowns.

The Augmented matrix is given by $[A:B] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & : & 0 \\ a_{21} & a_{22} & \dots & a_{2n} & : & 0 \\ \vdots & \vdots & \ddots & \vdots & : & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & : & 0 \end{bmatrix}$

$\rightarrow$ If $\rho(A) = \rho[A:B] = n$ the system is consistent and will have trivial soln $x_1 = x_2 = x_3 = \dots = x_n = 0$

$\rightarrow$ If $\rho(A) = \rho[A:B] = r < n$ the system will have non trivial soln and $(n-r)$ variables can be chosen arbitrarily

($\downarrow$
infinite no of solns)

Ex 1.1.1g :- $x + y + 3z = 0$, $4x + 3y + \lambda z = 0$, $2x + y + 2z = 0$

Solution of a system of non-homo eqn's

① Gauss Elimination Method :- This method is illustrated by considering a system of 3 eqns in 3 unknown

Consider the system of eqn's
$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{cases} \quad \text{--- (1)}$$

① is equivalent to the matrix form $AX = B$ where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

This method aims in reducing the co-eff matrix A to an upper (Δ) triangular matrix.

consider the Augmented matrix $[A:B] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & : & b_1 \\ a_{21} & a_{22} & a_{23} & : & b_2 \\ a_{31} & a_{32} & a_{33} & : & b_3 \end{bmatrix}$

Working rule :- We use the element a_{11} to make a_{21} & a_{31} to 0 by elementary row operation
 $\rightarrow$ ——— / ——— a_{22} to make a_{32} to zero ——— / ———

$\therefore$ The linear eqn is of the form
$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{33}x_3 &= b_3 \end{aligned}$$

Solve the following system of eqn by Gauss elimination method :-

①
$$\begin{cases} 4x_1 + x_2 + x_3 = 4 \\ x_1 + 4x_2 - 2x_3 = 4 \\ 3x_1 + 2x_2 - 4x_3 = 6 \end{cases} \quad \text{Sol: Given the system of eqns can be written in the matrix form as } AX = B$$

 $R_1 \leftrightarrow R_2 \quad \therefore \underline{x_1 = 1, x_2 = 1/2, x_3 = -1/2}$

②
$$\begin{cases} x_1 + x_2 + x_3 + 4x_4 = -6 \\ x_1 + 2x_2 + x_3 + x_4 = 12 \\ x_1 + x_2 + 6x_3 + x_4 = -5 \\ 5x_1 + x_2 + x_3 + x_4 = 4 \end{cases} \quad \begin{cases} x_1 = 1 \\ x_2 = 2 \\ x_3 = -1 \\ x_4 = -2 \end{cases}$$

③
$$\begin{cases} 2x + y + 4z = 12 \\ 4x + 11y - z = 33 \\ 8x - 3y + 2z = 20 \end{cases}$$

-! Gauss Jordan Elimination Method :-

This method can be regarded as the modification of Gauss Elimination Method. This method aims in reducing the co-efficient matrix 'A' to the diagonal Matrix.

Working rule:- Make use of the element a_{11} to make a_{21} & a_{31} to 0 by E.R.O. operation.

$$\begin{array}{rcl} & & a_{22} \rightarrow a_{12} \& a_{22} \text{ to zero} \\ & & a_{33} \rightarrow a_{13} \& a_{33} \text{ to zero} \end{array}$$

The linear eqn becomes $a_{11}x_1 = b_1$, $a_{22}x_2 = b_2$, $a_{33}x_3 = b_3$

Solving these we get x_1, x_2, x_3 values

EX'S :- (1)
$$\begin{cases} 2x + y + z = 10 \\ 3x + 2y + 3z = 18 \\ x + 4y + 9z = 16 \end{cases} \begin{cases} x = 7 \\ y = -9 \\ z = 5 \end{cases}$$
 Solve the following system of eqns by Gauss Jordan method

(2)
$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 2 \\ 2x_1 - x_2 + 2x_3 - x_4 = -5 \\ 3x_1 + 2x_2 + 3x_3 + 4x_4 = 7 \\ x_1 - 2x_2 - 3x_3 + 2x_4 = 5 \end{cases} \begin{cases} x_1 = 0 \\ x_2 = 1 \\ x_3 = -1 \\ x_4 = 2 \end{cases}$$

$$\begin{cases} x + 4y - z = -5 \\ x + y - 6z = -12 \\ 3x - y - z = 4 \end{cases}$$

Solve the following system of eqn by (a) Gauss elimination method
(b) Gauss Jordan method.

(1)
$$\begin{cases} x + y + z = 9 \\ x - 2y + 3z = 8 \\ 2x + y - z = 3 \end{cases} \begin{cases} x = 2 \\ y = 3 \\ z = 4 \end{cases}$$

(2)
$$\begin{cases} 2x + 5y + 7z = 52 \\ 2x + y - z = 0 \\ x + y + z = 9 \end{cases}$$

Soln:- The Augmented Matrix of the system is

$$[A:B] = \begin{bmatrix} 2 & 5 & 7 & : & 52 \\ 2 & 1 & -1 & : & 0 \\ 1 & 1 & 1 & : & 9 \end{bmatrix} \quad R_1 \leftrightarrow R_2$$

$$(a) [A:B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 2 & -1 & -1 & : & 0 \\ 2 & 5 & 7 & : & 52 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -3 & -3 & : & -18 \\ 0 & 3 & 5 & : & 34 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 + 3R_2 \end{array} \Rightarrow \sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -3 & -3 & : & -18 \\ 0 & 0 & -4 & : & -20 \end{bmatrix}$$

$\therefore$ The system of eqn becomes $x + y + z = 9$
 $\rightarrow x = 1$ $3y + z = -18$ $3y + 18 - z = -18 \Rightarrow 3y = -36 \Rightarrow y = -12$
 $x + y + z = 9 \Rightarrow x = 9 - y - z = 9 - (-12) - 3 = 18$
 $y + 3z = -18 \Rightarrow y = -18 - 3z = -18 - 15 = -33 \Rightarrow y = -33$
 $4z = 20 \Rightarrow z = 5$

$$(b) [A:B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 2 & 1 & -1 & : & 0 \\ 2 & 5 & 7 & : & 52 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -1 & -3 & : & -18 \\ 0 & 3 & 5 & : & 34 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 + 3R_2 \\ R_1 \rightarrow R_1 + R_2 \end{array} \Rightarrow \sim \begin{bmatrix} 1 & 0 & -2 & : & -9 \\ 0 & -1 & -3 & : & -18 \\ 0 & 0 & -4 & : & -20 \end{bmatrix} \begin{array}{l} R_2 \rightarrow -R_2 \\ R_3 \rightarrow -\frac{1}{4}R_3 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & -2 & : & -9 \\ 0 & 1 & 3 & : & 18 \\ 0 & 0 & 1 & : & 5 \end{bmatrix} \begin{array}{l} R_1 \rightarrow R_1 + 2R_3 \\ R_2 \rightarrow R_2 - 3R_3 \end{array} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & : & 1 \\ 0 & 1 & 0 & : & 3 \\ 0 & 0 & 1 & : & 5 \end{bmatrix}$$

$\therefore$ The system of eqn becomes $x = 1$
 $y = 3$
 $z = 5$

Gauss - Siedal Iterative method :-

This method is a numerical iterative method giving an approximate soln [whereas previous methods are exact soln]

This method is illustrated by considering a system of 3 eqns in 3 unknowns. i.e.

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

In this method the diagonal elements should be dominant
i.e. $|a_{11}| > |a_{12}| + |a_{13}|$; $|a_{22}| > |a_{21}| + |a_{23}|$; $|a_{33}| > |a_{31}| + |a_{32}|$

We start with the initial approximation i.e.

$$X = (x_1^{(0)}, x_2^{(0)}, x_3^{(0)}) = (0, 0, 0)$$

First approximation is

$$x_1^{(1)} = \frac{1}{a_{11}} [b_1 - a_{12}x_2^{(0)} - a_{13}x_3^{(0)}] = \frac{b_1}{a_{11}}$$

$$x_2^{(1)} = \frac{1}{a_{22}} [b_2 - a_{21}x_1^{(1)} - a_{23}x_3^{(0)}] =$$

$$x_3^{(1)} = \frac{1}{a_{33}} [b_3 - a_{31}x_1^{(1)} - a_{32}x_2^{(1)}] =$$

ii/y
2nd approx : $x_1^{(2)} = \frac{1}{a_{11}} [b_1 - a_{12}x_2^{(1)} - a_{13}x_3^{(1)}]$

$$x_2^{(2)} = \frac{1}{a_{22}} [b_2 - a_{21}x_1^{(2)} - a_{23}x_3^{(1)}]$$

$$x_3^{(2)} = \frac{1}{a_{33}} [b_3 - a_{31}x_1^{(2)} - a_{32}x_2^{(2)}]$$

iii/y others till the desired value.

Note:- If the given equations are not diagonally dominant then we have to make a rearrangement in the given eqn's to satisfy the above requirement.

Solve the following system of eqns by Gauss-Seidel method given

$$10x + y + z = 12; \quad x + 10y + z = 12; \quad x + y + 10z = 12$$

Soln:- Clearly the given eqns are diagonally dominant

Let us consider the initial approximation $X = (x^{(0)}, y^{(0)}, z^{(0)})$
 $= (0, 0, 0)$

1st approximation:-

Given eqns can be written in the form.

$$x^{(1)} = \frac{1}{10} (12 - x^{(0)} - y^{(0)}) = 1.2$$

$$y^{(1)} = \frac{1}{10} (12 - x^{(1)} - z^{(0)}) = 1.08$$

$$z^{(1)} = \frac{1}{10} (12 - x^{(1)} - y^{(1)}) = 0.972$$

2nd approx: $x^{(2)} = \frac{1}{10} (12 - 1.2 - 1.08) = 0.9948$

$$y^{(2)} = \frac{1}{10} (12 - 0.9948 - 0.972) = 1.0033$$

$$z^{(2)} = \frac{1}{10} (12 - 0.9948 - 1.0033) = 1.0002$$

3rd: $x^{(3)} = \frac{1}{10} (12 - 0.9948 - 1.0033) = 0.9996 \approx$

$$y^{(3)} = \frac{1}{10} (12 - 0.9996 - 1.0002) = 1.0000 \approx 1$$

$$z^{(3)} = \frac{1}{10} (12 - 0.9996 - 1) \approx 1$$

$\therefore \underline{x = 1; y = 1; z = 1.}$

2] $20x + y - 2z = 17$

$$3x + 20y - z = -18$$

$$2x - 3y + 20z = 25$$

$$x = 1; y = -1; z = 1$$

3] $x + y + 54z = 110$

$$27x + 6y - z = 85$$

$$6x + 15y + 8z = 72$$

$$x = 2.425; \quad y = 3.572$$

$$z = 1.925$$

not diagonally

dominant.

interchange.

$$7x + 52y + 13z = 104$$

$$3x + 8y + 29z = 71$$

$$83x + 11y - 4z = 95$$

$$x = 1.0579$$

$$y = 1.3672$$

$$z = 1.9617$$

Eigen values & Eigen vector of a square matrix

Let A be a square matrix of order n & consider the matrix of the form $A - \lambda I$, where ' λ ' is the scalar & I is the $n \times n$ id matrix. Then the eqn $|A - \lambda I| = 0$ is called the ch eqn of a square matrix of order n .

The roots of this ch eqn is called ch roots or eigen values of matrix A . They are denoted by $\lambda_1, \lambda_2, \dots, \lambda_n$.

Corresponding to each ch root λ we have a corresponding non-zero vector ' X ' which satisfies the eqn $[A - \lambda I]X = 0$.

These non-zero vectors are called ch roots ~~eqn roots~~ $\odot$ eigen values.

For each value of ' λ ' there will be an eigen vector $X \neq 0$ which is called as a characteristic vector $\odot$ Eigen vector.

Properties of Eigen values & eigen vectors:-

- ① Sum of the eigen values of a square matrix is equal to the trace (sum of the ^{principal} diagonal elements) of the matrix.
- ② Product of the eigen values of a square matrix is equal to the determinant of the matrix.
- ③ The eigen vector X of a matrix is not unique.
- ④ If $X_1 = (x_1, y_1, z_1)$, $X_2 = (x_2, y_2, z_2)$ then X_1, X_2 are called orthogonal vectors if $X_1 \cdot X_2 = x_1 x_2 + y_1 y_2 + z_1 z_2 = 0$.
- ⑤ Norm of a vector $X = (x, y, z)$ denoted by $\|X\|$ is equal to $\sqrt{x^2 + y^2 + z^2} = K$ (say). Then $(\frac{x}{K}, \frac{y}{K}, \frac{z}{K})$ is called the normalized vector.
- ⑥ Eigen vectors of a symmetric matrix corresponding to different eigen values are orthogonal.

Note:- The ch eqn of a 3rd order square matrix A can be obtained without expanding $|A - \lambda I| = 0$ by the following rule.

$$\lambda^3 - (\sum d) \lambda^2 + (\sum m_d) \lambda - |A| = 0, \text{ where}$$

$\sum d$ = Sum of the diagonal elements of A

$\sum m_d$ = Sum of the minors of the diagonal elements of A

Next Ex, $\sum d = 8 + 7 + 3 = 18$, $\sum m_d = 5 + 20 + 20 = 45$, $|A| = 0$

Substituting in the rule we get $\lambda^3 - 18\lambda^2 + 45\lambda = 0$.

Verification of Property: ① Sum of all eigen values $= 0 + 3 + 15 = 18 = \text{trace of } A = 18$

② Product of all eigen values $= 0$, $|A| = 0$

③ $X_1 \cdot X_2 = 2 + 2 - 4 = 0$, $X_2 \cdot X_3 = 4 - 2 - 2 = 0$, $X_3 \cdot X_1 = 2 - 4 + 2 = 0$.

$\therefore$ Eigen vectors X_1, X_2, X_3 are orthogonal.

④ $\|X_1\| = \sqrt{1+4+4} = 3$, $\|X_2\| = 3$, $\|X_3\| = 3 = K$.

normalized eigen vector of X_1, X_2, X_3 are resp $(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$, $(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3})$, $(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3})$

Orthogonal matrix $P = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ or $\frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ orthogonal matrix

Orthogonal matrix $= \begin{bmatrix} \frac{x_1}{\|x_1\|} & \frac{x_2}{\|x_2\|} & \frac{x_3}{\|x_3\|} \end{bmatrix}$

Eigen values & Eigen vectors :-

Let 'A' be a square matrix of order 'n'. Now consider the matrix of the form $A - \lambda I$, where λ is the scalar & I is the identity matrix of order 'n'. Then the eqn $|A - \lambda I| = 0$ is called the characteristic eqn of the square matrix 'A'.

The roots of this ch eqn are called ch roots (or) eigen values of matrix A. They are denoted by $\lambda_1, \lambda_2, \dots, \lambda_n$.

Corresponding to each ch root λ we have a corresponding non-zero vector 'X' which satisfies the eqn. $[A - \lambda I]X = 0$.

These non-zero vectors are called ch roots (or) eigen values.

Ex:- $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ find eigen eqn & eigen values.

Soln The ch eqn of A is $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0$.

$$= (8-\lambda)[(7-\lambda)(3-\lambda) - 16] + 6[(-6)(3-\lambda) + 8] + 2[24 - 2(7-\lambda)] = 0.$$

$$= \lambda^3 - 18\lambda^2 + 45\lambda = 0$$

$$= \lambda(\lambda^2 - 18\lambda + 45) = 0$$

$$\Rightarrow \lambda = 0, \lambda^2 - 18\lambda + 45 = 0.$$

$$\lambda = 0, \lambda = 3, 15$$

$\therefore \lambda = 0, 3, 15$ are the eigen values of A.

Now we form the system of eqn as

$$[A - \lambda I]X = 0.$$

where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$(x+a)(x+b)(x+c) = x^3 + (a+b+c)x^2 + (ab+bc+ca)x + abc$$

$$(x+a)(x+b) = x^2 + (a+b)x + ab$$

For solving homogeneous eqn we have rule of cross relation in the form

$$\frac{x}{1} = \frac{-4}{1} = \frac{-2}{1} = k$$

→ Find the ch roots & the corr eigen vector of the matrix

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \quad \lambda = 1, 2, 3$$

case(i): Let $\lambda = 1 \Rightarrow x + z = 0, y = 0, x + z = 0 \therefore x = -z$

$\therefore z = 1$ is arbitrarily chosen. for our convn the $\underline{x = -1}$

$\therefore (x, y, z) = (-1, 0, 1)^T = X_1$ is the eigen vector corresponding to $\underline{\lambda = 1}$

case(ii): Let $\lambda = 2, \Rightarrow z = 0, 0 = 0, x = 0 \therefore x = 0 \& y = 0$ can be chosen

arbitrarily say $y = 1$

$\therefore X_2 = (0, 1, 0)$ is the eigen vect to $\underline{\lambda = 2}$

case(iii) Let $\lambda = 3 \Rightarrow -x + z = 0, -y = 0, x - z = 0 \therefore x = z \& y = 0$

Let us choose $x = z = 1$ (arbitrary) $\therefore X_3 = (1, 0, 1)^T$ is the eigen vect to $\underline{\lambda = 3}$

$$\rightarrow A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \quad \lambda = -3, -3, 5$$

$X_1 = (3k_1 - 2k_2, k_2, k_1)$ with $\lambda = -3$

$X_2 = (1, 2, -1)$ with $\lambda = 5$

$$\rightarrow A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \quad \lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$\lambda = 2, 2, 8$

$$2 \begin{array}{ccc|c} 1 & -12 & 36 & -32 \\ 0 & 2 & -20 & 32 \\ 0 & -10 & 16 & 0 \end{array}$$

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix} \quad \lambda^3 - 7\lambda^2 + 8\lambda = 0$$

$$\lambda = -2, 3, 6$$

$$-2 \left[\begin{array}{ccc|c} 1 & -7 & 0 & 36 \\ 0 & -2 & 18 & -56 \\ 1 & -9 & 18 & 0 \end{array} \right] \quad (5)$$

case(i):- Let $\lambda = -2$, $3x + y + 3z = 0$
 $x + 2y + z = 0$
 $3x + y + 3z = 0$

$$\frac{x}{20} = \frac{-y}{0} = \frac{z}{-20} = (1, 0, -1)$$

$\therefore X = (1, 0, -1)^T$ is the eigen vector corresponding to $\lambda = -2$,

case(ii):- Let $\lambda = 3$, $-2x + y + 3z = 0$
 $x + 2y + z = 0$
 $3x + y - 8z = 0$

$$\frac{x}{-5} = \frac{-y}{-5} = \frac{z}{-5} \Rightarrow (-1, 1, -1)^T \text{ with } \lambda = 3$$

case(iii):- $\lambda = 6$, $-5x + y + 3z = 0$
 $x - y + z = 0$
 $3x + y - 5z = 0$

$$\frac{x}{4} = \frac{-y}{-8} = \frac{z}{4} \Rightarrow (1, 2, 1)^T \text{ with } \lambda = 6$$

→ S.T the eigen values of the following matrix are all equal. Find the corresponding eigen vector.

$$A = \begin{bmatrix} -3 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}$$

$\lambda = 1, 1, 1$. All the eigen values are equal

$X = (3, -1, -1)^T$ is the eigen vector corresponding to the coincident eigen value $\lambda = 1$

Q1

Rayleigh's Power Method :-

This is an iterative method to determine the numerically largest eigen value (dominant eigen value) & the corresponding eigen vector of a square matrix. This method is known as Power method.

Working rule :- Let A be a square matrix of order n & λ be its dominant eigen value. We first choose some column vector $x^{(0)}$ as the initial approximated eigen vector.

$$y^{(1)} = Ax^{(0)} \text{ \& re write in the form } y^{(1)} = \lambda^{(1)} x^{(1)}$$

where $\lambda^{(1)}$ is a real (or) complex no & $x^{(1)}$ is a column matrix

Then $\lambda^{(1)}$ is the first approximation to λ & $x^{(1)}$ is the first approximation to the corresponding eigen vector.

Next we find $y^{(2)} = Ax^{(1)}$ & re write in the form $y^{(2)} = \lambda^{(2)} x^{(2)}$

Then $y^{(2)}$ is the second approximation to λ & $x^{(2)}$ is the 2nd approximation to the corresponding eigen vector.

Ex's Find the largest eigen value & the corresponding eigen vector of the matrix. $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ by power method.

Soln :- Since initial approximation is not given, let us take initial approximation to the eigen vector corresponding to the required dominant eigen value of A is.

$$x^{(0)} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Using power method find the dominant eigen value & the corresponding eigen vector of the matrix A

Page No. 42

$$A = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \text{ taking } \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T \text{ as the initial eigen vector.}$$

[upto 3rd iteration]

Soln: let $x^{(0)} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ be the initial eigen vector.

$$\text{Then } y^{(1)} = A x^{(0)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 25 \\ 1 \\ 2 \end{bmatrix} = 25 \begin{bmatrix} 1 \\ 0.04 \\ 0.08 \end{bmatrix} = \lambda^{(1)} x^{(1)}$$

$$y^{(2)} = A x^{(1)} = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 0.04 \\ 0.08 \end{bmatrix} = \begin{bmatrix} 25.2 \\ 1.12 \\ 1.68 \end{bmatrix}$$
$$= 25.2 \begin{bmatrix} 1 \\ 0.04 \\ 0.07 \end{bmatrix} = \lambda^{(2)} x^{(2)}$$

$$y^{(3)} = A x^{(2)} = \begin{bmatrix} 25.8 \\ 1.12 \\ 1.72 \end{bmatrix} = 25.8 \begin{bmatrix} 1 \\ 0.04 \\ 0.07 \end{bmatrix} = \lambda^{(3)} x^{(3)}$$

Hence we conclude that the numerically largest eigen value is 25.8 & the corresponding eigen vector is

$$\begin{bmatrix} 1 \\ 0.04 \\ 0.07 \end{bmatrix}$$

Find the numerically largest eigen value & the corresponding eigen vector of the matrix

(16)
Page No. 43

$$A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \text{ by taking the initial approximation to the eigen vector as } \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T$$

(7 iterations)

Soln: Let $x^{(0)} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T$ be the initial eigen vector.

$$y^{(1)} = Ax^{(0)} = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ -2 \end{bmatrix} = 4 \begin{bmatrix} 1.0 \\ 0.5 \\ -0.5 \end{bmatrix} = \lambda^{(1)} x^{(1)}$$

$$y^{(2)} = Ax^{(1)} = \begin{bmatrix} 5 \\ 4 \\ -4 \end{bmatrix} = 5 \begin{bmatrix} 1.0 \\ 0.8 \\ -0.8 \end{bmatrix} = \lambda^{(2)} x^{(2)}$$

$$y^{(3)} = Ax^{(2)} = \begin{bmatrix} 5.6 \\ 5.208 \\ -5.208 \end{bmatrix} = 5.6 \begin{bmatrix} 1 \\ 0.93 \\ -0.93 \end{bmatrix} = \lambda^{(3)} x^{(3)}$$

$$y^{(4)} = Ax^{(3)} = \begin{bmatrix} 5.86 \\ 5.742 \\ -5.742 \end{bmatrix} = 5.86 \begin{bmatrix} 1.0 \\ 0.98 \\ -0.98 \end{bmatrix} = \lambda^{(4)} x^{(4)}$$

$$y^{(5)} = Ax^{(4)} = \begin{bmatrix} 5.96 \\ 5.900 \\ -5.900 \end{bmatrix} = \begin{bmatrix} 1.0 \\ 0.99 \\ -0.99 \end{bmatrix} = \lambda^{(5)} x^{(5)}$$

$$y^{(6)} = Ax^{(5)} = \begin{bmatrix} 5.98 \\ 5.98 \\ -5.98 \end{bmatrix} = 5.98 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \lambda^{(6)} x^{(6)}$$

Thus the dominant eigen value is 5.98 & the corresponding eigen vector is $\begin{bmatrix} 1 & 1 & -1 \end{bmatrix}^T$

Ex:-

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & 1 \\ 2 & -1 & 3 \end{bmatrix} \quad \text{initial } \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T$$